

MCANet: shared-weight-based MultiheadCrossAttention network for drug–target interaction prediction

Jilong Bian, Xi Zhang, Xiyang Zhang, Dali Xu and Guohua Wang

Corresponding author: Guohua Wang, College of information and Computer Engineering, Northeast Forestry University, 150040, Harbin, China.

E-mail: ghwang@nefu.edu.cn

Abstract

Accurate and effective drug–target interaction (DTI) prediction can greatly shorten the drug development lifecycle and reduce the cost of drug development. In the deep-learning-based paradigm for predicting DTI, robust drug and protein feature representations and their interaction features play a key role in improving the accuracy of DTI prediction. Additionally, the class imbalance problem and the overfitting problem in the drug–target dataset can also affect the prediction accuracy, and reducing the consumption of computational resources and speeding up the training process are also critical considerations. In this paper, we propose shared-weight-based MultiheadCrossAttention, a precise and concise attention mechanism that can establish the association between target and drug, making our models more accurate and faster. Then, we use the cross-attention mechanism to construct two models: MCANet and MCANet-B. In MCANet, the cross-attention mechanism is used to extract the interaction features between drugs and proteins for improving the feature representation ability of drugs and proteins, and the PolyLoss loss function is applied to alleviate the overfitting problem and the class imbalance problem in the drug–target dataset. In MCANet-B, the robustness of the model is improved by combining multiple MCANet models and prediction accuracy further increases. We train and evaluate our proposed methods on six public drug–target datasets and achieve state-of-the-art results. In comparison with other baselines, MCANet saves considerable computational resources while maintaining accuracy in the leading position; however, MCANet-B greatly improves prediction accuracy by combining multiple models while maintaining a balance between computational resource consumption and prediction accuracy.

Keywords: drug–target interaction, shared-weight-based MultiheadCrossAttention, ensemble model, deep learning

INTRODUCTION

Drug development and drug discovery is an important research field in biomedicine [1], and drug development companies invest billions of dollars on drug research every year [2]. Drug–target interactions (DTIs) are crucial for drug development [3]. Typically, extensive chemical and biological studies in the lab are performed by researchers to identify DTIs. Accurate prediction of DTIs can narrow down the scope of experimental screening, speed up drug development, lower research expenditures and it can even aid in the discovery of hidden functions of drugs and proteins.

Deep learning and machine learning have achieved superior performance in areas such as natural language processing and computer vision [4–7]. It has also achieved unquestionable performance in the field of bioinformatics [8]. Meanwhile, sequencing technology has become increasingly advanced over the past few years. As a result, the cost of sequencing is gradually decreasing, and the amount of sequencing data is growing exponentially. One is encouraged to mine valuable information hidden beneath these data using machine learning and deep learning algorithms

to advance biotechnology research. Nowadays, there is a large amount of data on DTIs, such as DrugBank, KIBA and Davis. Many approaches based on machine learning and deep learning have been developed to predict DTI using these data.

Faulon et al. [9] takes molecular signature and reaction signature as input to the kernel function of Support Vector Machines (SVM) to predict DTI. Meng et al. [10] combined Bi-gram probabilities (BIGP), Position Specific Scoring Matrix (PSSM) and Principal Component Analysis (PCA) with Relevance Vector Machine (RVM) to predict DTI. These structure-based methods work well for predicting DTIs for known structures, but poorly for unknown structures. Luo et al. [11] developed a quantitative structure–activity relationship (QSAR) model of the 5-HT_{1A} receptor, and used k-nearest neighbor algorithm (KNN), random forest algorithm and SVM algorithm to predict the drug–receptor interaction. Random forest [12] is also employed to predict DTI by Wang et al. [13], in which drug and target descriptors based on molecular vibrations are calculated and features extracted using the Boruta algorithm are taken as input to the random forest algorithm to

Jilong Bian is a lecturer at the College of Information and Computer Engineering, Northeast Forestry University. His main research interests include computer vision, deep learning and bioinformatics.

Xi Zhang is a master student at the College of Information and Computer Engineering, Northeast Forestry University. His main research interests include deep learning, natural language processing and bioinformatics.

Xiyang Zhang is an associate professor at the College of Information and Computer Engineering, Northeast Forestry University. Her main research interests include database technology, image retrieval and bioinformatics.

Dali Xu is a lecturer at the College of Information and Computer Engineering, Northeast Forestry University. Her main research interests include digital image processing and bioinformatics.

Guohua Wang is a professor at the College of Information and Computer Engineering, Northeast Forestry University. He is also a Principal Investigator at Key Laboratory of Tree Genetics and Breeding, Northeast Forestry University. His research interests are bioinformatics, machine learning and algorithms.

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perform prediction. However, the accuracy of results is affected since random forest algorithm is not robust to noise and hand-engineered features are insufficiently expressive.

Deep learning can automatically learn from training data through backward propagation of the features, which are insensitive to noise in data, and does not need to design features manually. Deep learning is used in a wide variety of fields since it outperforms traditional machine learning algorithms in terms of accuracy and efficiency. Nowadays, deep learning is adopted extensively in DTI prediction. Related research has shown that deep learning performs exceptionally well at identifying DTIs [14]. Lee et al. [15] proposed DeepConv-DTI, which employs a one-dimensional convolutional neural network (1D-CNN) to extract features from amino acid sequences of proteins, the extended connectivity fingerprints (ECFP) algorithm [16] to compute drug features and then a fully connected network (FCN) to predict DTI from these features. This method uses a convolutional network and an FCN to extract drug and protein features, but ignores interaction features of protein–drug pairs. Huang et al. [17] proposed MolTrans, a model that can create interpretable drug–target binding maps and predict DTI from the binding maps. In their study, features extracted by Transformer from SMILES sequences of drugs and amino acid sequences of proteins are mapped to a 2D binding map, which is then used as input to two-dimensional convolutional neural networks (2D-CNN) and FCNs to make predictions. This method utilizes only the interaction features of substructures of proteins and drugs but ignores their original features. Zhao et al. [18] proposed HyperAttentionDTI, in which an attention mechanism called HyperAttention is added to improve the feature expressiveness. Although this method considers the original features of proteins and drugs and their interaction features, HyperAttention has high spatio-temporal complexity and its effect is limited.

In addition to CNN and Transformer, there are many methods for predicting DTI based on graph neural network (GNN). In GNN model, proteins and drug molecules are expressed as graphs to represent the spatial and structural relationships of atoms and amino acids, and then features are extracted from the graphs to predict DTI. Tsubaki et al. [19] used GNN to extract features from compounds and CNN to extract features from protein sequences to predict their interactions. Zhao et al. [20] constructed a graph of drug–protein pairs (DPPs) from known drug–protein interactions, with DPP as the node and the edges inferred from known drug–protein interactions, drug–drug interactions and protein–protein interactions. Each DPP feature is extracted from the DPP network using GCN, and Deep Neural Network takes the DPP features as input to make classification. In addition, Wang et al. [21] proposed to use multiple views to capture the features of DPPs in topological and semantic networks, which improved the generalization ability of the model. However, the model has a lot of feature engineering. In order to solve this problem, SGCL-DTI [22] built an end-to-end DTI prediction model, learning the embedding of drugs and target proteins in heterogeneous networks through meta-path and attention mechanism. At the same time, for multiple views, the supervised graph co-contrastive learning is innovatively proposed, which provides a new prospect for DTI prediction research. As in [19], BACPI [23] used GNN and CNN to extract features, but the difference is that these features were enhanced using an attention mechanism. Wu et al. [24] developed a novel graph architecture called BridgeDPI, in which a set of hyper-nodes is constructed to connect all proteins and drugs. For each protein–drug pair, a graph is built from the set of hyper-nodes and the protein–drug pair. The graph is then fed into GNN to generate an

embedding of the protein–drug pair that is used to predict their interaction. Though GNNs are better at detecting relationships, the disadvantages of GCN are that it is inflexible, transductive and difficult to scale to large-scale networks, and slow to converge.

In other studies, features are extracted using a combination of approaches. Hua et al. [25] proposed CPInformer, which combines features extracted using GCN with functional class fingerprints (FCFP) of chemical molecules to form compound features and extracts features from protein molecules using CNN. Then, self-attention and sparse-attention from Informer [26] are used to optimize the features of chemical molecules and protein molecules, respectively. Finally, LeNet [27] is used to map the feature matrices to classification labels. Similar to MolTrans, this method ignores the original features of proteins. IIFDTI [28] proposed by Cheng et al. utilizes the encoder–decoder to extract the interaction features from DPPs and employs CNN and graph attention network (GAT) to extract the features of proteins and drugs, respectively. Then these features are concatenated and passed through MLP to make predictions. Although the transformer is used to extract interaction features, its structure is complex and often requires a large amount of training data to reach convergence. Wang et al. [29] came up with MCN-CPI, which takes as input not only compound molecular fingerprints and protein molecular sequences but also information about the binding site between the protein and the compound. It then uses 3D-CNN to extract binding site features in three-dimensional space, and these features are concatenated to make predictions. The method can only perform DTI prediction for proteins with 3D structures, which narrows down its application. Lee and Nam [30] proposed a model named HoTS consisting of two models: one for predicting binding regions (BRs) and the other for predicting DTIs. The BR model is pre-trained by taking compound-grid features as input. Then the pre-trained compound-grid features are fed into the DTI model to predict DTIs. Its prediction performance is sensitive to the quality of the features extracted in the pre-training phase. Such methods generally have more complex models with a large number of parameters. In the case of small amounts of data, underfitting might occur, and the organic combination of interaction features and initial features is also not taken into account.

Inspired by HyperAttentionDTI [18] and previous studies [15, 17], we propose an end-to-end neural network model for DTI prediction, namely MCANet. MCANet first learns the low-dimensional representation features from the sequences of proteins and drugs through two parallel CNN blocks. Then, our proposed shared-weight-based MultiheadCrossAttention module is utilized to learn the protein–drug interaction representations from their low-dimensional ones. Different from previous attentions, the shared-weight-based MultiheadCrossAttention computes attended protein feature first using protein as query and drug as key and value, and then computes attended drug feature using drug as query and protein as key and value. Since MCANet is a lightweight network model, we can make an ensemble of multiple trained MCANets generated during the k-fold cross-validation process, namely MCANet-B, and achieve a large improvement in prediction performance. In comparison with other SOTA baselines, our proposed methods show excellent performance on public drug–target datasets.

METHODS

The overall architecture of our proposed MCANet is shown Figure 1. It consists of four parts: two embedding layers for mapping drug sequences and protein sequences to embedding

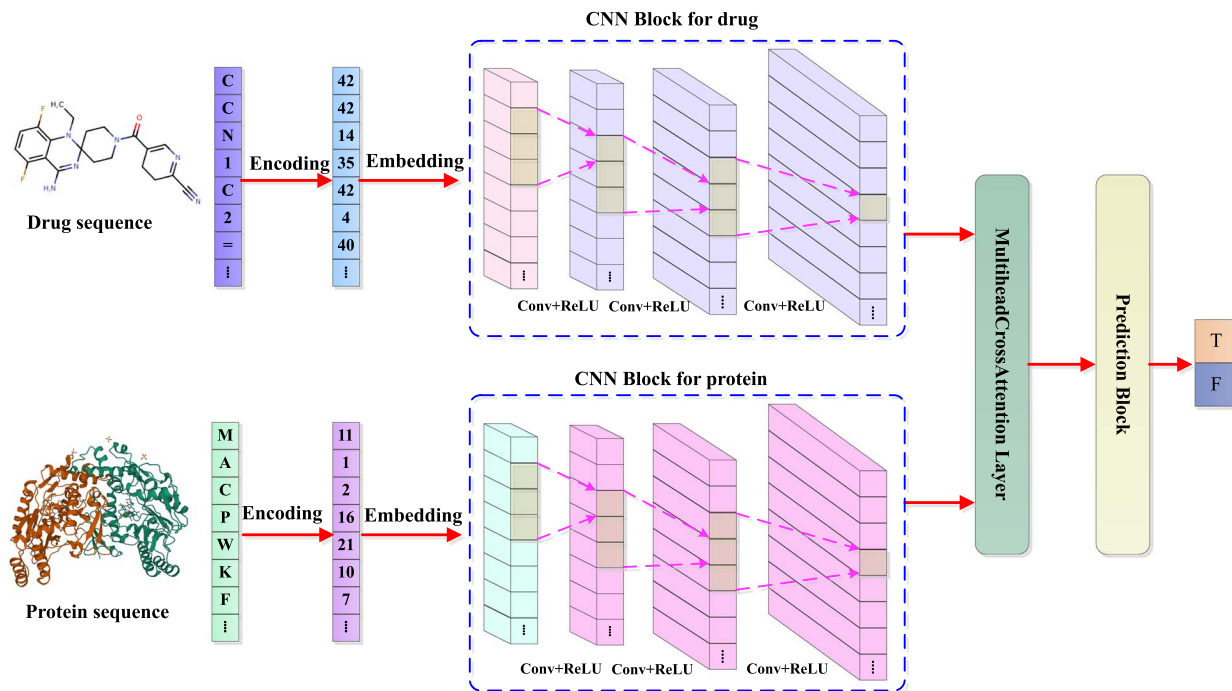


Figure 1. The overall architecture of our proposed MCANet. MCANet encodes input sequences through an Embedding layer, extracts features with CNN, then employs a shared-weight-based MultiheadCrossAttention module to extract the interaction features of drug–target pairs and finally gives predicted labels using a prediction block.

feature vectors; two CNN blocks for feature extraction; a shared-weight-based MultiheadCrossAttention block for extracting interaction features between drugs and proteins; and a prediction block for classification. In this proposed network model, digital codes for drugs and proteins are fed into the embedding layers to output embedding feature vectors, and they are taken as input to the CNN block for drugs and the CNN block for proteins, respectively, to generate representation features. The shared-weight-based MultiheadCrossAttention layer takes these representation features as input and mixes them to extract interaction features between drugs and proteins. Finally, these interaction features are utilized by the prediction block to predict DTI. The pipeline of MCANet is shown in Algorithm 1. We will provide more details about the modules in Algorithm 1 in the following sections.

Algorithm 1 The pipeline of MCANet

Input: D_{seq}, P_{seq} ▷ drug and protein sequences
 1: $D_{embed} \leftarrow \text{EmbeddingLayer1}(D_{seq})$
 2: $P_{embed} \leftarrow \text{EmbeddingLayer2}(P_{seq})$
 3: $D_{cnn} \leftarrow \text{CNNBlock1}(D_{embed})$
 4: $P_{cnn} \leftarrow \text{CNNBlock2}(P_{embed})$
 5: $D_f, P_f \leftarrow \text{MultiheadCrossAttention}(D_{cnn}, P_{cnn})$
 6: $lable \leftarrow \text{PredictBlock}(D_f, P_f)$
Output: $lable$ ▷ prediction of DTI

Embedding Layer

In the DTI prediction task, protein and drug sequences are first encoded into vector representations as input to the network model. We encode the tokens in SMILES sequences of drugs as integers ranging from 1 to 64 and the ones in amino acid sequences as integers ranging from 1 to 24. Then, one end of the encoded sequences is padded using 0's to keep the length of the sequences fixed. Finally, the encoded drug and protein

sequences of fixed length are fed into the embedding layers to generate a protein embedding matrix $P_{embed} \in \mathbb{R}^{L_p \times d_{embed}}$ and a drug embedding matrix $D_{embed} \in \mathbb{R}^{L_d \times d_{embed}}$, where L_p is the length of protein sequences, L_d is the length of drug sequences and d_{embed} is the channel dimension.

CNN Block

A convolution layer can be used to extract representation features by moving a fixed-size kernel across an image or sequence of data. The advantage of the convolution layer is that it can not only extract spatially associated information but also induce common features across an image or sequence of data. In our proposed model, we use two CNN blocks to extract the drug and protein features. Since drugs and proteins belong to different feature spaces, these two CNN blocks do not share weights. Figure 1 shows these two CNN blocks, each of which consists of three convolutions each followed by a ReLU activation function.

Shared-weight-based MultiheadCrossAttention

Attention mechanism has a wide application in natural language processing and computer vision due to its superior performance. Attention mechanism is adopted in Transformer to achieve excellent results in machine translation. After that, attention mechanisms attracted wide attention and were applied to many other areas. Attention module consists of Query vectors, Key vectors and Value vectors. It computes correlation coefficients between a query vector and key vectors, which serve as the weights of value vectors. The larger the weight is, the more attention it gets. However, self-attention mechanism determines the relationship between individual tokens within a sequence. In this paper, we propose a shared-weight-based MultiheadCrossAttention module to calculate cross attention between drug and protein to capture their interrelationships, which is beneficial to improving their feature representation.

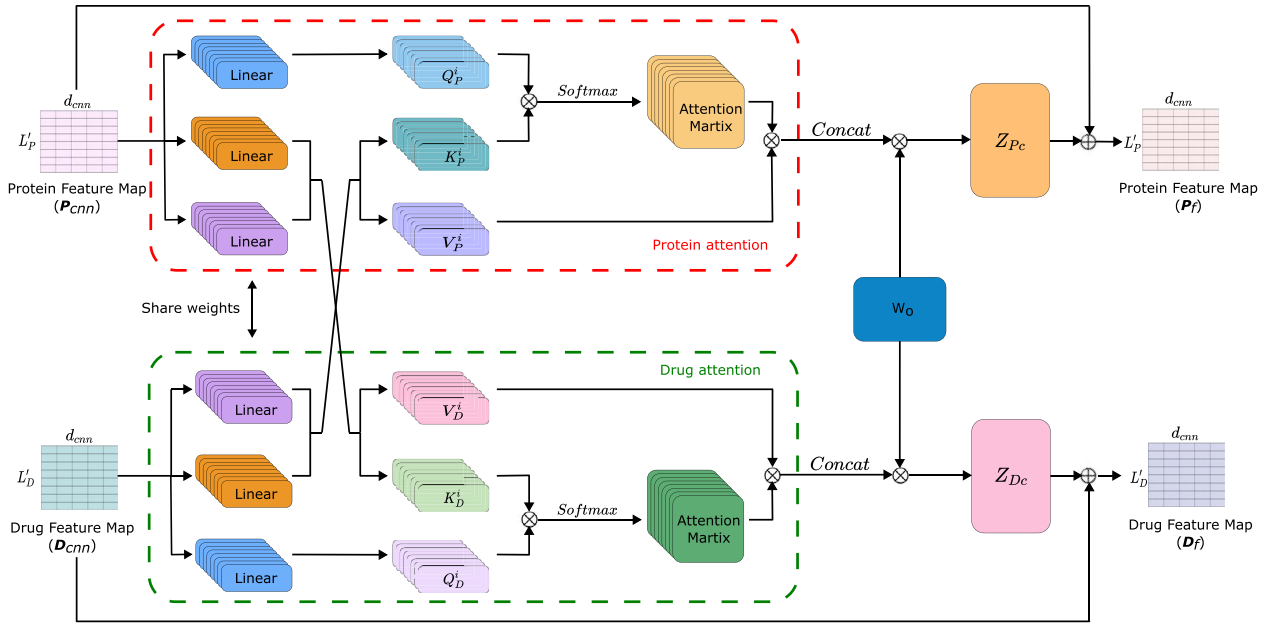


Figure 2. Diagram of MultiheadCrossAttention module. Linear layers with shared weights are used to map the features of drugs and proteins to queries, keys and values. This is followed by the exchange of keys and values between the protein attention submodule and the drug attention submodule. Then multi-head protein attention and multi-head drug attention are performed to generate the attended features Z_{Pc} and Z_{Dc} . Finally, these attended features are added to their initial features to generate final feature maps.

The architecture of the shared-weight-based Multihead-CrossAttention module is shown in Figure 2. It consists of protein attention and drug attention, both of which share the same weights. The drug attention component first passes a drug feature map $D_{cnn} \in \mathbb{R}^{L_D \times d_{cnn}}$ through a linear layer to compute a drug query vector $Q_D^i \in \mathbb{R}^{L_D \times d_{head}}$, and then takes a protein feature map $P_{cnn} \in \mathbb{R}^{L_P \times d_{cnn}}$ to compute a drug key vector $K_D^i \in \mathbb{R}^{L_P \times d_{head}}$ and a drug value vector $V_D^i \in \mathbb{R}^{L_P \times d_{head}}$ using two additional linear layers. The queries, keys and values for the drug are given by

$$\begin{cases} Q_D^i = D_{cnn} \times W_q^i, \\ K_D^i = P_{cnn} \times W_k^i, \\ V_D^i = P_{cnn} \times W_v^i, \end{cases} \quad i = 1, 2, \dots, heads \quad (1)$$

where $W_q^i, W_k^i, W_v^i \in \mathbb{R}^{d_{cnn} \times d_{head}}$ are the weight matrices of three linear layers, $d_{head} = d_{cnn}/heads$ is channel dimension and $heads$ is the number of attention heads.

The protein attention component follows a similar process to the drug attention component, and both attention components share these three linear layers. The protein attention component feeds a protein feature map into a linear layer to calculate a protein query vector $Q_P^i \in \mathbb{R}^{L_P \times d_{head}}$, and then takes a drug feature map to generate a protein key vector $K_P^i \in \mathbb{R}^{L_D \times d_{head}}$ and a protein value vector $V_P^i \in \mathbb{R}^{L_D \times d_{head}}$. The queries, keys and values for the protein are given by

$$\begin{cases} Q_P^i = P_{cnn} \times W_q^i, \\ K_P^i = D_{cnn} \times W_k^i, \\ V_P^i = D_{cnn} \times W_v^i, \end{cases} \quad i = 1, 2, \dots, heads \quad (2)$$

where the weight matrices $W_q^i, W_k^i, W_v^i \in \mathbb{R}^{d_{cnn} \times d_{head}}$ are shared by the protein attention and the drug attention.

Using a softmax function, the drug and protein attention matrices are computed as follows:

$$\begin{cases} S_D^i = \text{softmax} \left(\frac{Q_D^i K_P^{iT}}{\sqrt{d_{k_D^i}}} \right), \\ S_P^i = \text{softmax} \left(\frac{Q_P^i K_D^{iT}}{\sqrt{d_{k_P^i}}} \right), \end{cases} \quad i = 1, 2, \dots, heads \quad (3)$$

where $d_{k_D^i} = d_{k_P^i} = d_{head}$ is the dimension of queries and keys.

The drug attention matrix for each head is multiplied by the drug value matrix for each head to obtain the attended drug feature map for each head. Then the attended drug feature maps of all attention heads are concatenated along the channel dimension and fed into a linear layer to obtain the final attended drug feature map $Z_{Dc} \in \mathbb{R}^{L_D \times d_{cnn}}$:

$$Z_{Dc} = \text{Concat}(S_D^i \times V_D^i) \times W_o \quad i = 1, 2, \dots, heads \quad (4)$$

where $W_o \in \mathbb{R}^{d_{cnn} \times d_{cnn}}$ is weight matrix.

The same operations are applied to the protein attention matrix S_P^i and the protein value matrix V_P^i to obtain the final attended protein feature map $Z_{Pc} \in \mathbb{R}^{L_P \times d_{cnn}}$:

$$Z_{Pc} = \text{Concat}(S_P^i \times V_P^i) \times W_o \quad i = 1, 2, \dots, heads \quad (5)$$

where Eq. 4 and Eq. 5 share weight matrix W_o .

Finally, the attended feature maps are combined with the original feature maps to obtain the final feature maps:

$$\begin{cases} D_f = 0.5Z_{Dc} + 0.5D_{cnn} \\ P_f = 0.5Z_{Pc} + 0.5P_{cnn} \end{cases}, \quad (6)$$

where $D_f \in \mathbb{R}^{L_D \times d_{cnn}}$ denotes the drug feature map and $P_f \in \mathbb{R}^{L_P \times d_{cnn}}$ denotes the protein feature map.

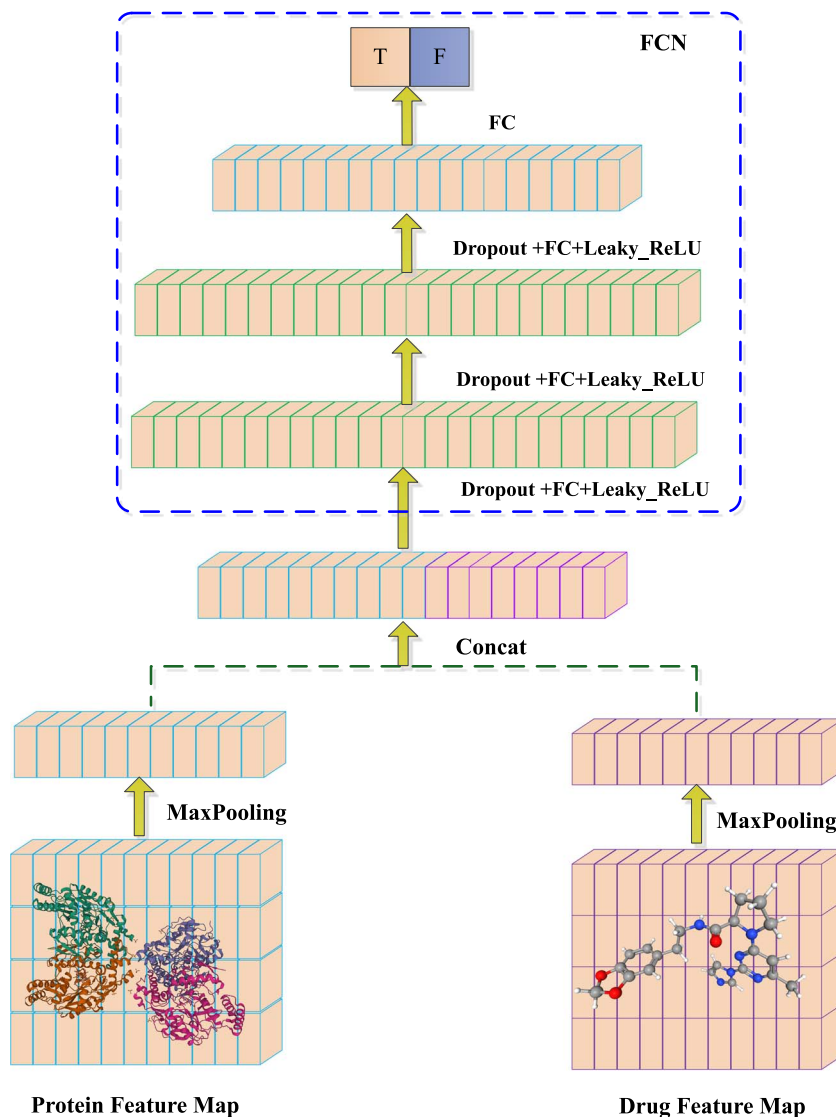


Figure 3. Diagram of prediction block. The prediction block first compresses 2D feature maps for proteins and drugs into 1D feature maps through MaxPolling layers. Then they are concatenated and fed into FCN for classification.

Analysis: Extracting the interaction features between amino acids in proteins and atoms in drug molecules is the most critical step in the process of predicting DTI. To this end, we propose a useful tool called shared-weight-based MultiheadCrossAttention for capturing the interaction features between drugs and proteins to improve the accuracy of prediction. Self-attention uses the same sequence to generate queries, keys and values to capture the interrelationships within the sequence. However, our proposed shared-weight-based MultiheadCrossAttention utilizes a drug molecule sequence and a protein sequence to generate queries, keys and values to capture their interrelationships. The attended feature maps are combined with the initial feature maps to produce mixed feature maps. Mixed feature map includes interaction features between drugs and proteins in addition to its own features. In our proposed MultiheadCrossAttention, the drug attention component and protein attention component share weights to extract the binding features between drugs and proteins, which frequently suggest potential interaction sites.

Prediction Block

As shown in Figure 3, the prediction block consists of two global maxpooling layers, a concatenation layer, and an FCN. The global

maxpooling layers downsample the protein feature map $P_f \in \mathbb{R}^{L_p \times d_{cm}}$ and the drug feature map $D_f \in \mathbb{R}^{L_d \times d_{cm}}$ to generate 1D feature vectors $P_{pooling} \in \mathbb{R}^{d_{cm}}$ and $D_{pooling} \in \mathbb{R}^{d_{cm}}$:

$$\begin{cases} D_{pooling} = \text{maxpooling}(D_f) \\ P_{pooling} = \text{maxpooling}(P_f) \end{cases} \quad (7)$$

Then, the drug and protein feature maps are concatenated:

$$V_C = \text{Concat}(D_{pooling}, P_{pooling}), \quad (8)$$

where $V_C \in \mathbb{R}^{2d_{cm} \times 1}$.

Finally, the concatenated feature V_C is fed into the FCN to make DTI predictions. The FCN consists of dropout layers, fully connected layers and leaky ReLU activation functions [31]. The dropout layer is used to avoid overfitting during training. The arrangement of the first three layers is the dropout layer, the fully connected layer and the leaky ReLU activation function. However, the last layer consists only of a fully connected layer.

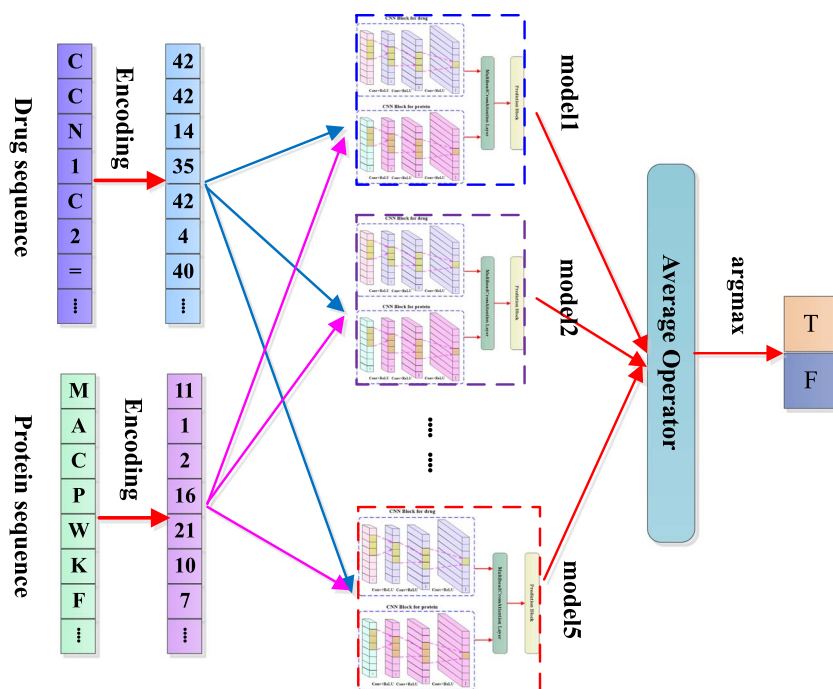


Figure 4. MCANet-B ensemble Model. MCANet-B consists of several MCANets with different weights, and their prediction probabilities are averaged to give predicted labels.

Ensemble Model

During the K-fold Cross-Validation, K models can be generated, and the training datasets used for these K models are not exactly the same, which results in different weights for trained models as well. Therefore, the models obtained during the cross-validation process can be combined to generate a larger model with better prediction accuracy. In the actual training process, we use a 5-fold cross-validation procedure. The dataset is first divided into two parts: the training dataset and the test dataset. Then the training dataset is again divided into five parts. The four folds of the data are used as training data to train the model. The remaining part of the data is used as validation data to evaluate the resulting model. In this way, we can obtain five trained models, which are combined to make a large model called MCANet-B, as shown in Figure 4. This approach can effectively improve prediction accuracy without increasing training resource consumption, but only at the expense of proper memory consumption during inference.

Loss function

Loss function is crucial for training neural network models in deep learning. It measures how close the predicted values of the neural network model is to its corresponding target values. Selecting a suitable loss function can greatly reduce training time and errors, as well as improve the generalization ability. Currently, the cross-entropy loss function is the most widely used loss function in classification problems:

$$CE_{loss} = -[t \log(p) + (1 - t) \log(1 - p)], \quad (9)$$

where p is the predicted probability and t is the corresponding target value.

The loss function is a log-likelihood function, whose advantage is that weights are updated quickly when the loss is large and slowly when the loss is small. In a binary classification task, the target value of negative examples is typically set to 0 and the one

of positive examples is set to 1. Therefore, the cross-entropy loss function can be reduced to

$$CE_{loss} = -\log P_t, \quad (10)$$

where P_t is the predicted probability of the target value.

Recently, Leng et al. [32] proposed the PolyLoss loss function defined as

$$\text{Poly-N} = -\log P_t + \sum_{i=1}^N \epsilon_i (1 - P_t)^i, \quad (11)$$

where ϵ_i is the perturbation term.

When $\epsilon_i = 0$, the PolyLoss degenerates into a binary cross-entropy loss function. The cross-entropy loss and focal loss are two special cases of PolyLoss. Therefore, Polyloss can mitigate the class imbalance. The PolyLoss has been demonstrated to obtain good performance in a variety of computer vision tasks. In order to strike a balance between performance and efficiency, we use Poly-1 as the loss function in our implementation:

$$\text{Poly-1} = -\log(P_t) + \epsilon_1(1 - P_t), \quad (12)$$

where ϵ_1 is set to 1.

EXPERIMENTS

Datasets

We first trained and evaluated our proposed methods using three publicly available datasets: Davis [33], KIBA [34] and DrugBank. Davis and KIBA are datasets including pairs of kinase-related proteins and drugs, in which we analyzed the similarity among drug molecules and the homology among proteins. As shown in Supplementary Material Fig.S1-Fig.S4, the similarity among drug molecules inside Davis and KIBA is low, but there are some

homologous proteins inside these two datasets. DrugBank is a comprehensive dataset including information on drugs and their targets. The Davis dataset consists of the interactions between 72 kinase inhibitors and 442 kinases, in which 68 drugs and 379 proteins are singled out to obtain 7320 positive examples and 18 452 negative examples. The KIBA dataset is the dataset on kinase inhibitor bioactivity (KIBA). It utilizes 2068 drugs and 225 proteins to make 22 154 positive examples and 94 196 negative examples. For DrugBank, we employed the dataset construction method used in HyperAttentionDTI [18] excluding very tiny drug molecules and those not recognized by RDKit [35] to make 17 511 positive examples and 17 511 negative examples from 4294 proteins and 6655 drugs.

As the main targets for drug action, proteins can typically be divided into Enzymes, GPCRs and Ion Channels and so on. Thus, we construct three more datasets on Enzymes, Ion channels and GPCRs to evaluate prediction performance across different target categories. Enzymes and Ion channels are collected from public databases: KEGG BRITE[36], DrugBank, SuperTarget[37] and BRENDA[38], which are available from <http://web.kuicr.kyoto-u.ac.jp/supp/yoshi/drugtarget/>. For Enzymes, we construct 2920 positive examples and 2920 negative examples from 660 proteins and 444 drugs. For Ion channels, 204 proteins and 210 drugs are used to generate 1475 positive examples and 1475 negative examples. GPCRs are collected from GLASS database[39], from which 296 proteins and 567 drugs are selected to produce 3098 positive examples and 3099 negative examples.

Experimental Settings

Hyper-Parameters

Proper hyperparameters such as learning rate, batch size, epoch and convolution kernel size can help a network model converge quickly and produce more accurate results. Since drug sequences are typically shorter than protein sequences, we use different hyperparameter configurations in the CNN blocks for proteins and drugs. In the CNN block for drugs, the kernel sizes of three convolutions are set to 4, 6 and 8, while in the CNN block for proteins, the kernel sizes of three convolutions are set to 4, 8 and 12. In the process of training, the AdamW optimizer is used to update the network weights. It has two hyperparameters: learning rate and weight decay, which are both set to $1e-4$. The batch size and the maximum number of epochs are set to 16 and 200, respectively. Early stopping is also employed to avoid overfitting. We use the same hyperparameters in all baselines to ensure that the comparisons are fair. For more details about the model parameters and architecture parameters, please refer to Table S1 in the supplementary material.

Baselines

DeepConv-DTI: DeepConv-DTI [15] consists of convolution layers and fully connected layers. It uses CNN to extract the feature maps of proteins and FCN to extract the feature maps of drugs, and then performs prediction.

MolTrans: MolTrans [17] utilizes the transformer, which is now making a splash in a variety of fields, to encode protein and drug sequences into feature embeddings, which are then used to create an interaction matrix. Finally, CNNs and FCNs are applied to the interaction matrix to make predictions.

TransformerCPI: In TransformerCPI [40], interactions between compounds and proteins are predicted using CNN, GCN and Transformer. Firstly, CNN is used to extract features from protein sequences, and GCN is used to extract features from drug sequences. Then the Transformer decoder is employed to convert

drug and protein features into inputs to FCN. Finally, FCN makes predictions.

HyperAttentionDTI: In HyperAttentionDTI [18], CNN serves as the backbone to extract drug and protein features, which are mixed to capture their interactions using HyperAttention mechanism. These mixed features are fed into FCN to perform DTI prediction.

Training settings and experimental Environment

The proposed method is implemented using Pytorch [41] and trained on the following hardware configuration:

- CPU: Intel i5-4460, 3.20GHz
- GPU: NVIDIA GTX1080Ti, 11GB
- Memory: 16GB DDR3

During training, the dataset is divided into the training dataset and the test dataset according to the 80–20% rule. Then the training dataset is again divided into five parts. Four of the folds are used as training data to train the model and the remaining part of the data is used as validation data to validate the model.

Results

Analysis of performance

To validate the performance of our proposed MCANet, we trained MCANet and the baselines on three datasets: DrugBank, Davis and KIBA, and a series of evaluation metrics such as accuracy, precision, recall, AUC and AUPR are used to evaluate them. During training, the same hyperparameters, such as learning rate, epoch number, etc., are utilized in order to fairly evaluate the performance of these methods.

Firstly, MCANet and the baselines are trained and evaluated on the DrugBank dataset, and the experimental results are presented in Table 1. The experimental results show that our proposed MCANet holds the top position on multiple evaluation metrics compared with the baselines. As compared with the best-performing baseline, HyperAttentionDTI, our proposed MCANet achieves improvements of 1.6% accuracy, 2.52% precision, 0.81% AUC and 2.01% AUPR. In order to improve prediction performance further, we propose MCANet-B, an ensemble model that achieves large improvements on all evaluation metrics, with improvements of 2.88% accuracy, 3.42% precision, 2.09% recall, 2.4% AUC and 2.39% AUPR. It has been shown that our proposed MCANet and MCANet-B can achieve a large prediction performance gain on the DrugBank dataset, demonstrating their feasibility and effectiveness.

Secondly, our proposed methods and the baselines are trained and evaluated on the Davis dataset. This is a more challenging dataset because the class imbalance in the dataset has an adverse impact on the performance of deep network models. For this reason, the weighted PolyLoss loss function is used to reduce the penalty of majority classes and increase the penalty of minority classes. The experimental results are shown in Table 2. Our proposed MCANet achieves an improvement of 0.64% accuracy, 0.31% recall, 0.46% AUC and 0.75% AUPR over the best-performing baseline, DeepConv-DTI. Our ensemble model MCANet-B has a large improvement on all metrics, with improvements of 2.86% accuracy, 4.8% precision, 2.4% recall, 2.77% AUC and 5.07% AUPR. The experimental results show that our proposed MCANet and MCANet-B can also achieve performance improvement on the challenging Davis dataset.

Finally, we train and evaluate our proposed methods and the baselines on the KIBA dataset. This is also an imbalance dataset similar to the Davis dataset, and its degree of imbalance

Table 1. Comparison results of the proposed models and baselines on the DrugBank dataset

Methods	Accuracy	Precision	Recall	AUC	AUPR
DeepConv-DTI	0.7950	0.7930	0.8045	0.8669	0.8628
MolTrans	0.7848	0.7701	0.8197	0.8598	0.8667
TransformerCPI	0.7823	0.7725	0.8075	0.8543	0.8585
HyperAttentionDTI	0.8100	0.7990	0.8290	0.8890	0.8844
MCANet	0.8260	0.8242	0.8274	0.8971	0.9045
MCANet-B	0.8548	0.8584	0.8483	0.9211	0.9284

Table 2. Comparison results of the proposed model and baselines on the Davis dataset

Methods	Accuracy	Precision	Recall	AUC	AUPR
DeepConv-DTI	0.8641	0.7785	0.7214	0.9183	0.8387
MolTrans	0.8361	0.6887	0.7636	0.8984	0.8026
TransformerCPI	0.8208	0.7047	0.6272	0.8645	0.7470
HyperAttentionDTI	0.8636	0.7602	0.7619	0.9210	0.8436
MCANet	0.8705	0.7754	0.7667	0.9256	0.8511
MCANet-B	0.8927	0.8265	0.7876	0.9487	0.8943

Table 3. Comparison results of the proposed model and baselines on the KIBA dataset

Methods	Accuracy	Precision	Recall	AUC	AUPR
DeepConv-DTI	0.8905	0.8052	0.6887	0.9067	0.8042
MolTrans	0.8891	0.7042	0.7353	0.9232	0.7949
TransformerCPI	0.8850	0.7200	0.6598	0.9070	0.7648
HyperAttentionDTI	0.8775	0.6730	0.7149	0.9161	0.7721
MCANet	0.8995	0.7352	0.7498	0.9313	0.8109
MCANet-B	0.9168	0.7889	0.7764	0.9504	0.8620

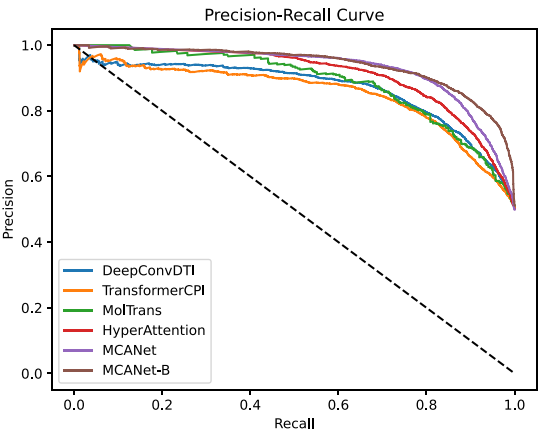


Figure 5. P-R curves of our proposed methods and other baselines on DrugBank.

is more severe and challenging than that of the Davis dataset. The experimental results are shown in Table 3. As compared with the best-performing baseline, DeepConv-DTI, our proposed MCANet improves accuracy, recall, AUC and AUPR by 0.9%, 1.45%, 0.81% and 0.67%, respectively. In addition, our ensemble model, MCANet-B, further improves accuracy, recall, AUC and AUPR by 2.63%, 4.11%, 2.72%, and 5.78%, respectively. The experimental results demonstrate that our proposed methods also work well on the more challenging KIBA dataset.

We plotted the Precision-Recall Curves of our proposed methods and other baselines on DrugBank in Figure 5, through which it is observed that our proposed MCANet and MCANet-B are better than other baselines in terms of prediction performance. We

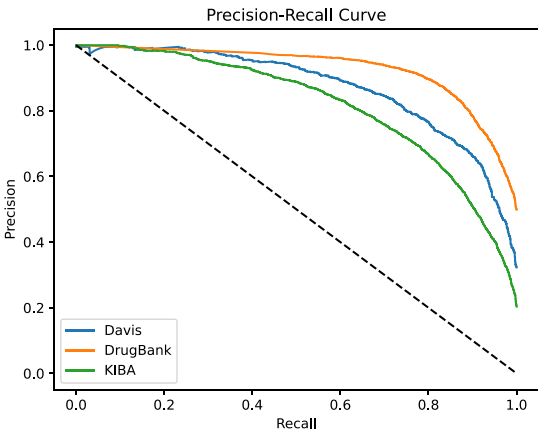


Figure 6. P-R curves of MCANet on DrugBank, Davis and KIBA.

also plotted the Precision-Recall Curves of MCANet on DrugBank, Davis and KIBA in Figure 6 to evaluate its performance across different datasets. It is observed that MCANet performs best on DrugBank. This is mainly because DrugBank is a balanced dataset, while the other two datasets are imbalanced datasets.

We again conduct experiments on three additional datasets: Enzymes, Ion channels and GPCRs, to evaluate prediction performance across different target categories. As shown in Table 4, our proposed MCANet and MCANet-B outperform other baselines on Enzymes, Ion channels, and they also outperform other baselines other than DeepConv-DTI on GPCRs. By and large, MCANet and MCANet-B have high prediction performance for Enzymes, Ion channels and GPCRs, with the highest prediction performance for

Table 4. Comparison of the proposed models and baselines on Enzymes, Ion channel and GPCRs

		DeepConvDTI	MolTrans	TransformerCPI	HyperAttentionDTI	MCANet	MCANet-B
Enzymes	Accuracy	0.7474	0.7271	0.5009	0.4512	0.7597	0.8106
	AUC	0.8017	0.7888	0.4835	0.4181	0.8039	0.8576
	AUPR	0.7568	0.7348	0.4905	0.4557	0.7301	0.7917
Ion channels	Accuracy	0.6475	0.7153	0.4670	0.3375	0.7285	0.7691
	AUC	0.6850	0.7166	0.4577	0.4524	0.7615	0.8203
	AUPR	0.6340	0.6421	0.4619	0.4602	0.7005	0.7535
GPCRs	Accuracy	0.6303	0.6055	0.5381	0.5860	0.6242	0.6486
	AUC	0.6724	0.6391	0.5669	0.6094	0.6662	0.6975
	AUPR	0.6403	0.6023	0.5410	0.5887	0.6149	0.6457

Table 5. Statistical comparison results

Paired methods	Accuracy	Precision	Recall	AUC	AUPR
MCANet VS DeepConvDTI	1.055E-04	0.873	5.030E-06	8.510E-07	1.865E-04
MCANet VS TransformerCPI	4.186E-07	1.276E-05	2.378E-05	5.722E-08	4.556E-07
MCANet VS HyperAttentionDTI	2.486E-06	1.901E-04	0.162	8.366E-04	9.189E-04
MCANet VS MolTrans	4.893E-06	5.036E-06	0.114	4.924E-06	4.721E-07

Enzymes. In particular, MCANet-B outperforms other baselines by a wide margin on these three datasets.

Statistical comparison

To further confirm the performance of the proposed MCANet, a paired T-test is used to make a statistical comparison between MCANet and other baselines on different evaluation metrics, such as Accuracy, Precision, Recall, etc. In the paired T-test, the null hypothesis (H_0) is that MCANet is not significantly greater than the chosen baseline on a given evaluation metric and the alternative hypothesis (H_1) is that MCANet is significantly greater than the chosen baseline on a given evaluation metric. Significance level is set at 0.05. If the observed P-value is lower than 0.05, then H_0 is rejected and H_1 is accepted, showing that MCANet is significantly greater than the chosen baseline on the given evaluation metric. MCANet is tested one by one against other baselines using the paired T-test for significant differences on each evaluation metric. We use k-fold cross validations on DAVIS, KIBA and DrugBank to gather the values for each evaluation metric and then conduct a paired T-test on them. Table 5 shows the statistical results for pair-wise comparison of MCANet and other baselines. Each row shows the P-values for different metrics. It can be observed from these statistical results that our proposed MCANet performs significantly better than other baselines in terms of Accuracy, AUC and AUPR. In addition, it performs significantly better than other baselines other than DeepConvDTI in terms of precision. Since MCANet-B is built on MCANet, MCANet-B can also perform significantly better than other baselines. Hence we can conclude that our proposed MCANet and MCANet-B have significantly better performance for most of the evaluation metrics.

Analysis of efficiency

In this section, we analyze the GPU memory consumption and model complexity of our proposed methods and the baselines. The GPU memory consumption is primarily composed of the network model and other processing components in the prediction method. It is measured by the GPU memory used during training. Model complexity is measured by the number of parameters and floating-point operations (FLOPs). The model training time is also collected to measure the time complexity

of the prediction methods. The experimental results are shown in Table 6. DeepConv-DTI has the least memory consumption, number of parameters and FLOPs during training. It is due to the fact that this method includes only the network module for extracting protein features, but not the network module for extracting drug molecule features, which are obtained by RDKit during the data preprocessing process. The proposed model MCANet ranks second in terms of memory consumption and FLOPs, using only 1345 M memory and 0.399 G FLOPs, while its accuracy is improved by 3.1% over the first-ranked DeepConv-DTI model. By comparison with HyperAttentionDTI, MCANet uses only one-fourth of its GPU memory and one-tenth of its FLOPs and achieves an improvement of 1.6% accuracy. In terms of training time, MCANet takes the least amount of time among all methods, only 8 h. The experimental results show that our proposed MCANet has better performance in terms of memory consumption, model complexity, time complexity and accuracy. Our proposed ensemble model, MCANet-B, further improves prediction accuracy at the cost of the number of parameters and computation cost. Since MCANet-B makes ensemble of models trained during K-fold cross-validation, no additional training cost and GPU memory are required. In comparison with MolTrans, MCANet-B reduces GPU memory consumption, model complexity and time complexity, while increases accuracy by 7%. Compared with the HyperAttentionDTI, MCANet-B only has a slight increase in the number of parameters but increases accuracy by 4.48%. In summary, MCANet strikes a balance between accuracy and efficiency, whereas MCANet-B achieves higher accuracy at the cost of the number of parameters and computation cost.

Ablation experiments

We conduct ablation experiments on the DrugBank dataset to evaluate the effectiveness of the MultiheadCrossAttention (MCA) and PolyLoss loss function used in our network model. We first remove the MultiheadCrossAttention module from MCANet to construct a variant model, short for CNN-PB-PolyLoss and compare it with MCANet to validate the effectiveness of the MultiheadCrossAttention mechanism. As shown in Table 7, the MultiheadCrossAttention improves accuracy, AUC and AUPR by 0.95%, 0.75% and 0.82%, respectively. Therefore, it can effectively

Table 6. Comparison of the proposed method and the baseline methods over GPU memory usage, training time, number of weights, accuracy and FLOPs on the DrugBank dataset.

Methods	Memory (MB)	Training time (hour)	Params (M)	FLOPs (G)	Accuracy
DeepConv-DTI	911	13.5	0.494	0.002	79.50%
MolTrans	7810	68.4	47.163	68.686	78.48%
TransformerCPI	3961	18.5	0.447	15.384	78.23%
HyperAttentionDTI	4393	33	2.303	4.714	81.00%
MCANet	1345	8	2.226	0.399	82.60%
MCANet-B	1345	-	11.13	1.993	85.48%

Table 7. Ablation experiments on DrugBank.

Methods	Accuracy	AUC	AUPR
CNN-PB-PolyLoss	0.8165	0.8895	0.8963
CNN-MCA-PB-CEloss	0.8120	0.8898	0.8962
MCANet(CNN-MCA-PB-PolyLoss)	0.8260	0.8971	0.9045

improve the model’s performance. As shown in Figure 7(A, B), MCANet consistently outperforms CNN-PB-PolyLoss on accuracy and AUPR throughout the training process, further demonstrating the effectiveness of the MultiheadCrossAttention. Then we replace the PolyLoss loss function of MCANet with the cross-entropy loss function (CEloss) to construct another model, called CNN-MCA-PB-CEloss and compare it with MCANet to verify the effectiveness of PolyLoss loss function. As shown in Table 7, the PolyLoss loss function improves accuracy, AUC and AUPR by 1.4%, 0.73% and 0.83%, respectively. Figure 7(C, D) shows that MCANet consistently outperforms CNN-MCA-PB-CEloss on accuracy and AUPR. The cross-entropy loss cannot prevent overfitting during the middle and late stages of training, which results in a decrease in accuracy, whereas the performance of MCANet using the PolyLoss steadily improves, demonstrating that the PolyLoss is also effective in improving the model’s performance.

Performance analysis of ensemble model

In this section, we discuss the impact of the number of network models combined in the ensemble model MCANet-B on performance. On the DrugBank dataset, we investigated the change in accuracy and AUPR of an ensemble model with different numbers of network models combined. The results are shown in Figure 8. MCANet is trained using 5-fold cross-validation, hence there are five trained models. When MCANet-B consists of only one model, it degenerates into MCANet, and as the number of models combined increases, its accuracy and AUPR increase as well. The experimental results show that the model’s performance can be effectively improved by a combination of multiple models. The number of models in the ensemble model is limited by GPU memory. Since MCANet occupies less GPU memory, the prediction performance can be increased by making an ensemble of multiple models.

Case Study

To demonstrate the effectiveness and generalization ability of our method, we randomly selected from the DrugBank dataset two target proteins and the drugs that interact with them. We then removed these proteins and drugs and used the remaining data as training data to predict them. We show in Table 8 the results of predicting drug–target pairs associated with proteins DRD5 (P21918) and HTR2A (P28223). Our model correctly predicted the

Table 8. Prediction result of protein DRD5 and HTR2A

Protein	Drug	Prediction	Real
P21918-DRD5	DB00800	True	True
	DB01624	True	True
	DB00714	True	True
	DB01038	True	True
	DB01186	True	True
	DB01200	True	True
	DB00589	True	True
	DB00248	True	True
	DB05271	True	True
	DB00246	True	True
	DB08584	False	False
	DB09040	False	False
	DB00590	False	False
	DB04519	False	False
	DB02411	False	False
	DB04628	False	False
	DB07934	False	False
	DB01099	False	False
	DB00712	False	False
	DB04824	True	False
Accuracy			95%
P28223-HTR2A	DB00247	True	True
	DB00370	True	True
	DB00434	True	True
	DB00656	True	True
	DB00924	True	True
	DB01149	True	True
	DB01224	True	True
	DB00246	True	True
	DB00334	True	True
	DB00363	True	True
	DB02858	False	False
	DB03453	False	False
	DB02405	False	False
	DB02894	False	False
	DB00547	False	False
	DB05676	False	False
	DB00307	True	False
	DB02235	False	False
	DB08709	True	False
	DB03820	False	False
Accuracy			90%

20 positive samples, but produced three false predictions among the 20 negative samples.

Once again, we randomly selected two drugs from the DrugBank dataset, as well as the target proteins that interact with them,

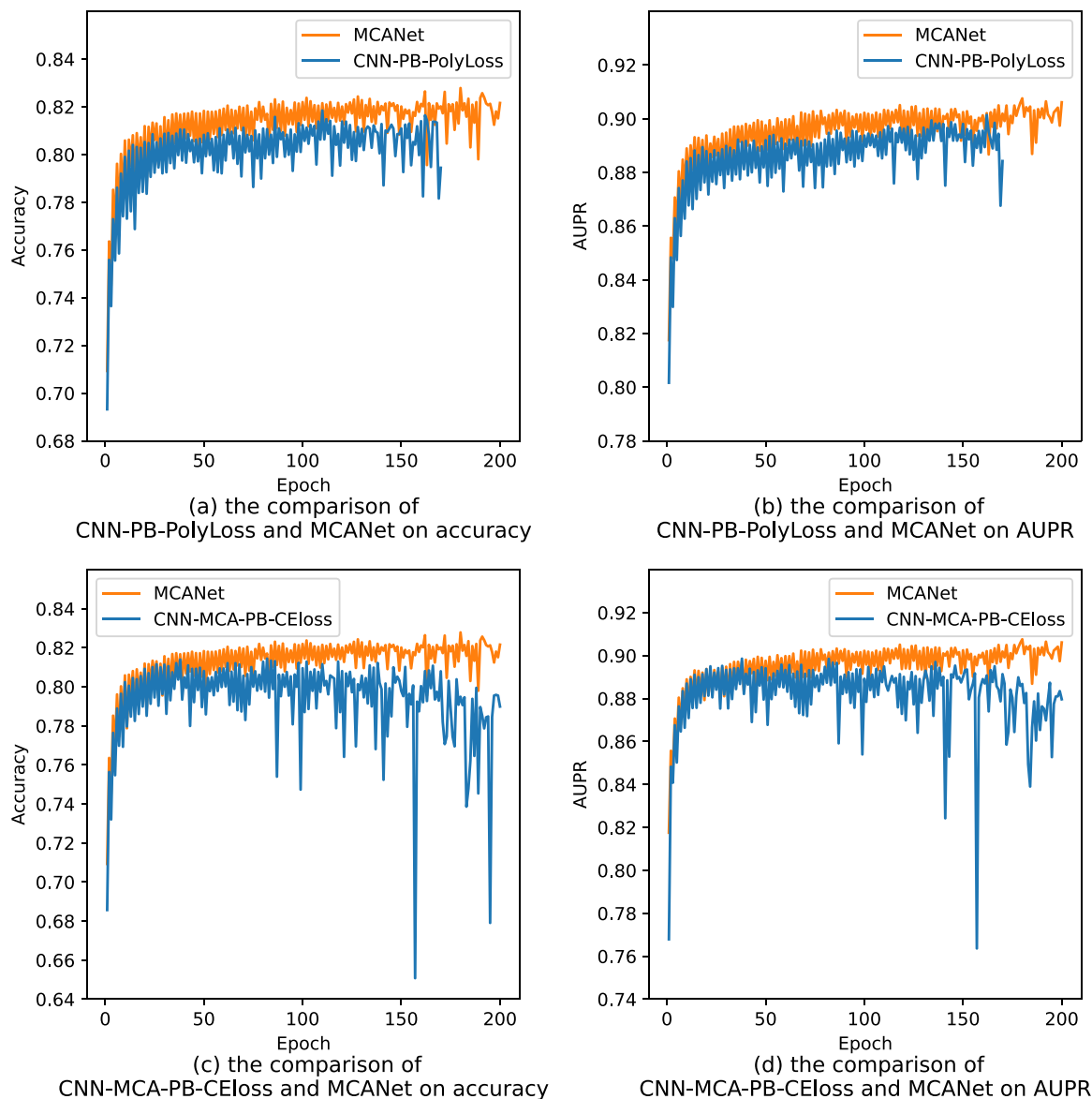


Figure 7. Comparison of CNN-PB-PolyLoss, CNN-MCA-PB-CEloss and MCANet on accuracy and AUPR. (A) the comparison of CNN-PB-PolyLoss and MCANet on accuracy. (B) the comparison of CNN-PB-PolyLoss and MCANet on AUPR. (C) the comparison of CNN-MCA-PB-CEloss and MCANet on accuracy. (D) the comparison of CNN-MCA-PB-CEloss and MCANet on AUPR.

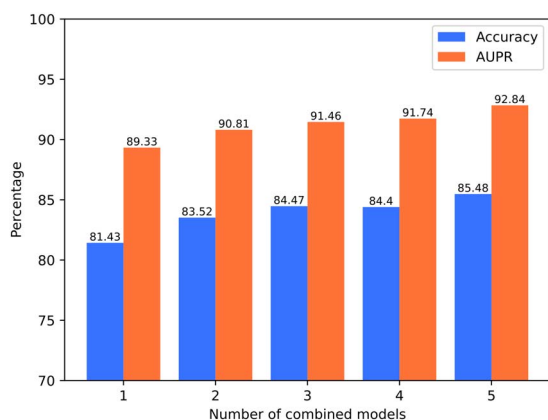


Figure 8. Accuracy and AUPR with different numbers of models combined in MCANet-B.

and removed them. Then, we used the remaining data to train our method to predict them. Sorafenib (DB00398) and Sunitinib (DB01268) are our two randomly selected drugs. Sorafenib is a targeted drug for the treatment of renal cell carcinoma and hepatocellular carcinoma by blocking multiple pathways and inhibiting blood vessel growth within the tumor. Sunitinib, a small molecule drug used to treat renal cell carcinoma and gastrointestinal stromal tumors, can interact with targets such as KIT and VEGFR2. Table 9 shows the results of interaction between these two drugs and multiple targets. For sorafenib (DB00398), we achieved a success rate of 90%, with one false prediction in 10 positive samples and one false prediction in 10 negative samples. For sunitinib (DB01268), we achieved a success rate of 95%, with one false prediction in 10 negative samples. The above experimental results show that our proposed method has good prediction and generalization ability in predicting drug-target pairs.

Table 9. Prediction result of drug Sorafenib and Sunitinib

Drug	Protein	Prediction	Real
DB00398-Sorafenib	P15056	True	True
	P04049	False	True
	P35916	True	True
	P35968	True	True
	P36888	True	True
	P09619	True	True
	P10721	True	True
	P11362	True	True
	P07949	True	True
	P17948	True	True
	O34450	False	False
	Q9GZT4	True	False
	Q27738	False	False
	P30920	False	False
	Q13002	False	False
	P21728	False	False
	P11229	False	False
	Q7RTX0	False	False
	O06796	False	False
	P35368	False	False
Accuracy			90%
DB01268-Sunitinib	P09619	True	True
	P17948	True	True
	P10721	True	True
	P35968	True	True
	P35916	True	True
	P36888	True	True
	P07333	True	True
	P16234	True	True
	P08581	True	True
	Q9P0Z9	False	False
	P02359	False	False
	P19099	False	False
	Q75Y35	False	False
	P13827	False	False
	P55210	True	False
	P00488	False	False
	P06149	False	False
	Q9NPC2	False	False
	P43099	False	False
Accuracy			95%

DISCUSSION

In this work, we developed a shared-weight-based Multihead-CrossAttention network, namely MCANet, to predict DTIs. In MCANet, we designed a lightweight MultiheadCrossAttention with shared weights to extract the interaction features of drug-target pairs. Experimental results showed that our proposed shared-weight-based MultiheadCrossAttention improved prediction performance by extracting the interaction features of drug-target pairs. This module is lightweight and has fewer parameters and FLOPs. Our interaction features are bidirectional, meaning that not only does the protein interact with the drug, but also the drug interacts with the protein, which fully expresses the interaction pattern between the protein and the drug and lends itself to improving the feature representation ability. To the best of our knowledge, this is the first attempt that uses bi-directional interaction features to improve feature representation. We conducted comprehensive experiments on

DrugBank, KIBA and Davis, showing that our proposed MCANet outperformed other baselines on most of the evaluation metrics. Meanwhile, statistical comparison was also performed on these datasets, indicating that MCANet is significantly better than other baselines for most of the evaluation metrics. In addition, we carried out experiments on Enzymes, Ion channels and GPCRs, demonstrating that MCANet works well for Enzymes from three different protein categories. The data size of the datasets for these three categories is small, but our methods work very well on such datasets, especially MCANet-B, which outperforms other baselines by a large margin, showing that our methods are more advantageous on small datasets.

From the above experimental results, we can draw the following conclusions:

- (I) the transformer-based methods (e.g. MolTrans and TransformerCPI) do not outperform CNN-based methods (e.g. DeepConv-DTI) in terms of prediction performance. The reason for this might be that the transformer-based method requires more training data to achieve better prediction performance.
- (II) The cross-attention mechanism effectively increased prediction performance, indicating that interaction features of drug-target pairs highlight the feature representation of BRs to generate more powerful feature representation.
- (III) MCANet-B makes an ensemble of the models generated during K-fold cross-validation to further increase prediction performance. This is because in the worst case, the prediction performance of the ensemble model will be equal to or greater than that of the single model with the lowest prediction performance in the ensemble model. Therefore, it is possible to combine as many models as possible without exceeding GPU memory limits to make high-quality predictions.

Most previous DTI studies take protein sequences as input to predict DTI. One obvious reason is that the 3D structure of proteins is more difficult to obtain than protein sequences. Thus, the number of the 3D structure of proteins is less than the number of protein sequences. However, for deep learning, a large amount of data is required to train a network model for reasonable prediction performance. Therefore many network models take sequence information as input. Our methods rely only on protein sequences and not on the 3D structure of proteins, so they can predict DTI for proteins without a 3D structure. This gives our model a wider applicability. However, our methods also have limitations. At present, there are two types of drugs: orthosteric drugs and allosteric drugs. Orthosteric drugs bind directly to the primary active sites, namely the orthosteric sites, causing inhibition or modification of the function of proteins. In contrast, allosteric drugs bind to secondary binding sites, namely the allosteric sites, causing changes in the conformation of the orthosteric sites to affect protein activity. However, our methods can only predict whether a drug molecule can bind to a protein; we are not able to distinguish between allosteric binding and orthosteric binding.

Though our proposed methods achieved large improvements in prediction performance, there still remains room for further improvement. Since our proposed methods are lightweight models, they can be extended by adding multiple types of inputs to further improve prediction performance. For proteins, statistical features such as K-mer features and PSSM can be used as inputs to the network model in addition to amino acid sequences; for drugs, MACCS fingerprint and PubChem fingerprint can be taken as inputs to the network model other than the SMILES string.

Additionally, hyperparameters for network models in the field of DTI are typically set manually according to the engineer's experience. Therefore, in the future we will employ neural architecture search (NAS) to determine the model's hyperparameters to further improve prediction performance.

CONCLUSION

We present an end-to-end, resource-efficient deep learning model, known as MCANet, which extracts features from the sequences of drug molecules and amino acids and performs DTI prediction using these features. This model uses the shared-weight-based MultiheadCrossAttention module to efficiently extract robust representation features between drugs and targets and employs the PolyLoss loss function to improve the overfitting problem and the class imbalance problem. Compared with the existing state-of-the-art methods that use only protein sequences and drug molecule sequences for DTI prediction, our proposed MCANet achieves leading performance on six different datasets. Additionally, we propose an ensemble model MCANet-B, which obtains a significant improvement in prediction performance by making an ensemble of multiple models generated during the k-fold cross-validation process.

Key Points

- We present shared-weight based MultiheadCrossAttention, which extracts the interaction features of drug-target pairs to highlight features of BRs for more powerful feature representation. Meanwhile, it is a generic cross-attention mechanism and can be easily extended to other applications.
- We propose a shared-weight-based MultiheadCrossAttention network, namely MCANet, for DTI prediction. Through the MultiheadCrossAttention with shared weights and the PolyLoss loss, MCANet outperforms other baselines by a wide margin on prediction performance across six public datasets. Furthermore, this model is lightweight, saving a considerable amount of computational resources.
- Since MCANet is a lightweight network model, we build an ensemble model, namely MCANet-B, by a combination of multiple MCANets with different weights and achieve a large performance improvement across six public datasets.

SUPPLEMENTARY DATA

Supplementary data are available online at <https://academic.oup.com/bib>.

AVAILABILITY

Codes and Datasets are available at <https://github.com/MrZQAQ/MCANet>

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